JAVA Talk 5. Atiyah class and Chern character

$$\begin{tikzcd}
\Delta \arrow[d, hook] \arrow[drr, hook, bend left, "\iota_\Delta"] & & \\
2\Delta \arrow[rr, hook, "\iota_{2\Delta}"] & & Y \times Y \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\
& Y & & Y & \mathcal{F} \arrow[l, dash]
\end{tikzcd}$$

Denote $\mathcal{O} = \mathcal{O}_{Y \times Y}$

$I = I_\Delta$

$j^1(\mathcal{F}) := Rp_{1,*}\left(\mathcal{O}/I^2 \otimes p_2^* \mathcal{F}\right)$

$= Rp_{1,*}\, \iota_{2\Delta,*}\, \iota_{2\Delta}^* \, p_2^* \mathcal{F}$

at: $$\begin{array}{ccccccccc} 0 & \longrightarrow & I/I^2 & \longrightarrow & \mathcal{O}/I^2 & \longrightarrow & \mathcal{O}/I & \longrightarrow & 0 \\ & & \| & & & & \| & & \\ & & \iota_{\Delta,*}\Omega_Y & & & & \iota_{\Delta,*}\mathcal{O}_Y & & \end{array}$$

$-\otimes p_2^*\mathcal{F}$: $$0 \to I/I^2 \otimes p_2^*\mathcal{F} \to \mathcal{O}/I^2 \otimes p_2^*\mathcal{F} \to \mathcal{O}/I \otimes p_2^*\mathcal{F} \to 0$$

$Rp_{1,*}$: $$\begin{array}{ccccccccc} 0 & \longrightarrow & \Omega_Y \otimes \mathcal{F} & \longrightarrow & Rp_{1,*}\left(\mathcal{O}/I^2 \otimes p_2^*\mathcal{F}\right) & \longrightarrow & \mathcal{F} & \longrightarrow & 0 \\ & & & & \|\text{def} & & & & \\ & & & & j^1(\mathcal{F}) & & & & \end{array}$$

one defines that

$$at_{\mathcal{F}} = A(\mathcal{F}) \in \mathrm{Ext}^1(\mathcal{F}, \Omega_Y \otimes \mathcal{F})$$

Rmk. 1). Since $I/I^2, \mathcal{O}/I^2, \mathcal{O}/I$ are supported on 2Δ

and $p_1 \circ \iota_{2\Delta}$ is finite,

$$Rp_{1,*}(I/I^2 \otimes -) = p_{1,*}(I/I^2 \otimes -).$$

2) $$\Omega_Y \xlongequal{\text{def}} \iota_\Delta^*(I/I^2) = p_{1,*}(I/I^2) \overset{1)}{=\!=} Rp_{1,*}(I/I^2)$$

$$\neq$$

$$\mathbb{L}_Y \xlongequal{\text{def}} L\iota_\Delta^*(I/I^2)$$

(cotangent complex)

Reason: I/I^2 is supported in Δ

$$\Rightarrow \exists\ \mathcal{E} \in \mathrm{Coh}(\Delta),\ I/I^2 = \iota_{\Delta,*}\mathcal{E}$$

$$\Rightarrow \iota_\Delta^*(I/I^2) = \iota_\Delta^*\iota_{\Delta,*}\mathcal{E}$$

$$\xlongequal[\text{embedding}]{\iota_\Delta \text{ is closed}} \mathcal{E} = p_{1,*}\iota_{\Delta,*}\mathcal{E} = p_{1,*}(I/I^2)$$

3) $$\mathcal{O}_\Delta = \iota_\Delta^*(\mathcal{O}/I^2) \neq p_{1,*}(\mathcal{O}/I^2) \overset{1)}{=\!=} Rp_{1,*}(\mathcal{O}/I^2)$$

$$\neq$$

$$L\iota_\Delta^*(\mathcal{O}/I^2)$$

E.g. When $Y = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$, $\operatorname{char} k \neq 2$,

$$\Omega_Y = k[\varepsilon]d\varepsilon/(2\varepsilon d\varepsilon) \cong k \cdot d\varepsilon$$

is not free.

$$R := \mathcal{O}_{Y\times Y} = k[\varepsilon_1, \varepsilon_1]/(\varepsilon_1^2, \varepsilon_2^2) \qquad \cong k^4$$

$$I = (\varepsilon_2 - \varepsilon_1) \overset{R\text{-mod}}{\cong} R/(\varepsilon_1 + \varepsilon_2) \cong k^{\oplus 2}$$

$$I^2 = (2\varepsilon_1 \varepsilon_2) \overset{R\text{-mod}}{\cong} R/(\varepsilon_1, \varepsilon_2) \cong k$$

$$I/I^2 = k(\varepsilon_2 - \varepsilon_1) \overset{R\text{-mod}}{\cong} R/(\varepsilon_1, \varepsilon_2) \cong k$$

$$\mathcal{O}/I^2 = \langle 1, \varepsilon_1, \varepsilon_2 \rangle \overset{R\text{-alg}}{\cong} R/(\varepsilon_1, \varepsilon_2) \cong k^3$$

$$p_{1,*}(I/I^2) = k$$

$$i_\Delta^*(I/I^2) = k[\varepsilon]/(\varepsilon^2) \otimes_R R/(\varepsilon_1, \varepsilon_2)$$

$$= k[\varepsilon]/(\varepsilon^2, \varepsilon, \varepsilon)$$

$$= k$$

$$L i_\Delta^*(I/I^2) = k[\varepsilon]/(\varepsilon^2) \otimes_R^L R/(\varepsilon_1, \varepsilon_2)$$

$$= \Big[\to \underset{-2}{k[\varepsilon]/(\varepsilon^2)} \xrightarrow{\binom{\varepsilon}{-\varepsilon}} \underset{-1}{k[\varepsilon]/(\varepsilon^2)^{\oplus 2}} \xrightarrow{(\varepsilon, \varepsilon)} \underset{0}{k[\varepsilon]/(\varepsilon^2)} \Big] \to$$

$$L^i i_\Delta^*(I/I^2) . \qquad k \qquad k^{\oplus 2} \qquad k$$

We use a free resolution of I/I^2.

$$0 \longrightarrow \underset{2}{R} \xrightarrow{\binom{\varepsilon_2}{-\varepsilon_1}} \underset{1}{R^{\oplus 2}} \xrightarrow{(\varepsilon_1, \varepsilon_2)} \underset{0}{R} \longrightarrow I/I^2 \longrightarrow 0$$

$$p_{1,*}(\mathcal{O}/I^2) = k[\varepsilon_1]/(\varepsilon_1^2)\cdot 1 \oplus k\cdot \varepsilon_2$$

$$\begin{aligned} l_\Delta^*(\mathcal{O}/I^2) &= k[\varepsilon]/(\varepsilon^2) \otimes_R R/(\varepsilon_1\varepsilon_2) \\ &= k[\varepsilon]/(\varepsilon^2, \varepsilon^2) \\ &= k[\varepsilon]/(\varepsilon^2) \end{aligned}$$

$$L^i l_\Delta^*(\mathcal{O}/I^2) = \begin{cases} k & \text{for } i>0 \\ k[\varepsilon]/(\varepsilon^2) & \text{for } i=0 \end{cases}$$

We use a free resolution of $\mathcal{O}/I^2$:

$$\begin{tikzcd}
\underset{3}{R^{\oplus 3}} \arrow[rr, "\phi_2", "\begin{pmatrix} \varepsilon_1 & -\varepsilon_2 & 0 \\ 0 & \varepsilon_1 & \varepsilon_2 \end{pmatrix}"'] \arrow[dr] & & \underset{2}{R^{\oplus 2}} \arrow[rr, "\phi_1", "(\varepsilon_1, \varepsilon_2)"'] \arrow[dr] & & \underset{1}{R} \arrow[r, "\varepsilon_1\varepsilon_2"] & \underset{0}{R} \arrow[r] & \mathcal{O}/I^2 \arrow[r] & 0 \\
& \left\langle \begin{pmatrix} \varepsilon_1 \\ 0 \end{pmatrix}, \begin{pmatrix} \varepsilon_2 \\ -\varepsilon_1 \end{pmatrix}, \begin{pmatrix} \varepsilon_1\varepsilon_2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ \varepsilon_2 \end{pmatrix}, \begin{pmatrix} 0 \\ \varepsilon_1\varepsilon_2 \end{pmatrix} \right\rangle \arrow[ur] & & \langle \varepsilon_1, \varepsilon_2, \varepsilon_1\varepsilon_2 \rangle \arrow[ur] & & & &
\end{tikzcd}$$

Where
$$\phi_i = \begin{pmatrix} \varepsilon_1 & (-1)^{i+1}\varepsilon_2 & & \\ & \varepsilon_1 & (-1)^i \varepsilon_2 & \\ & & \ddots & \ddots \end{pmatrix}_{i\times(i+1)}$$